

Notes: Flanagan (JF, 2025)

The Value of Bank Lending

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1 Big picture

- **Question.** How much value do banks create when they lend, after adjusting loan cash flows for public-market risk?
- **Core measurement.** The paper estimates risk-adjusted profit (RAP) for syndicated loan cash flows by comparing realized loan cash flows to a replicating portfolio of public securities with similar risk.
- **Main result.** Bank loans generate about \$0.032 of RAP per \$1 invested, or roughly 177 basis points in annualized gross risk-adjusted returns.
- **Economic mechanism.** Higher returns are concentrated in lending to financially constrained borrowers, where theory predicts banks have more scope to screen, monitor, and mitigate financing frictions.
- **Net returns.** Once staff compensation and other noninterest expenses are deducted, shareholders earn nearly zero net risk-adjusted returns.

2 Data and measurement

The paper constructs realized cash flows for syndicated term loans to U.S. public firms from Dealscan, incorporating prepayments, modifications, refinancings, defaults, recoveries, and fees.

Interpretation. The key data contribution is that the analysis uses realized loan cash flows rather than promised spreads.

3 Models and empirical specifications

3.1 Economic framework

$$\psi - \gamma \geq 0, \quad \theta_{bond} - (\theta_{loan} + \psi) \geq 0.$$

Interpretation. The bank must capture enough value to cover lending-service costs, and the borrower takes a bank loan only if the remaining surplus is positive.

3.2 Risk adjustment

$$E_t[M_{t,t+h}R_{t+h}^k] = 1, \\ X_{t+h}^i = a_{t+h} + b_h R_{t+h}^k + e_{t+h}^i.$$

Loan benchmark funds incorporate the outstanding loan balance:

$$\tilde{R}_{t+h}^i = R_{t+h} z_t^i, \quad E_t[M_{t,t+h} \tilde{R}_{t+h}^i] = z_t^i.$$

The main regression is

$$X_{t+h}^i = a_{t+h} + \sum_k \left[b_k \tilde{F}_{t+h}^{i,k} + c_{kh} \tilde{G}_{t+h}^{i,k} \right] + e_{t+h}^i. \\ RAP = E[PV(\text{loan cash flows})] - 1, \quad \psi = RAP/WAL.$$

Interpretation. The benchmark funds price public-market risks. A positive RAP means the price of a replicating public portfolio exceeds the \$1 cost of investing in the loan portfolio.

4 Identification and empirical design

The key assumption is that the public benchmark funds span the priced risk in loan cash flows. The corporate bond placebo test supports this assumption because the method finds near-zero risk-adjusted returns for bonds whose market prices are known.

5 Tables and figures with detailed captions and expanded interpretations

5.1 Figure 1: Loan Cash Flow Construction; Table I: Summary Statistics

Read: Figure 1 explains the cash-flow construction, while Table I describes the economic scale and main features of the loan sample.

- The figure separates scheduled payments from realized cash flows. Realized cash flows include scheduled principal and interest, fees, prepayments, renegotiations, refinancings, defaults, and recoveries.
- Table I shows that the loans are large private-credit contracts with relatively short realized lives and frequent ex post events.
- The key object is therefore not a promised spread at origination, but the full realized payoff stream to the lender.

Detailed interpretation: Figure 1 and Table I establish why the paper’s cash-flow reconstruction is essential. Syndicated loans are not static bonds: principal balances decline, borrowers refinance, contracts are amended, and default recoveries may arrive at different times. A spread-only analysis would miss these events. Table I also shows that the sample is economically important and representative of the market segment where banks’ private lending services are most visible.

How it fits the paper: This is the foundation for the rest of the paper. The subsequent risk-adjusted-return estimates are only meaningful because the loan cash flows have been measured as realized cash flows rather than assumed contractual payments.

Economic message. The measurement contribution begins with recognizing that bank loans are dynamic contracts. Bank value is earned through the realized sequence of loan cash flows, not only through the stated coupon.

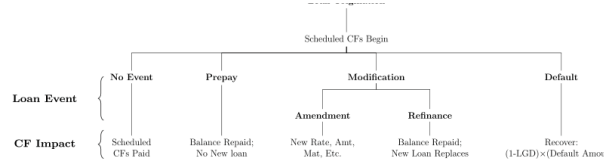


Figure 1. Loan cash flow construction. This figure shows the basic steps for constructing loan cash flows. If a loan does not default and is not repaid early, I simply take the scheduled cash flow from Dealscan as the realized cash flows. If a loan is prepaid, modified, or defaults, I adjust the scheduled cash flow accordingly to generate the realized cash flows.

This relationship indicates that, by revealed preference, the present value

Figure 1: Loan Cash Flow Construction. This figure shows how the paper builds the realized loan cash-flow series from contract terms and subsequent loan events. The important takeaway is that the empirical payoff includes actual repayments, prepayments, fees, refinancing outcomes, default payments, and recoveries rather than only scheduled payments.

5.2 Figure 2: Average Loan Cash Flow Distribution; Figure 3: Loan Balance Amortization

Read: These figures describe the timing of loan cash flows and the decline in outstanding balance after origination.

- Figure 2 shows that realized cash flows are front-loaded relative to a fixed long-term bond because loans amortize, prepay, or refinance.
- Figure 3 shows that loan balances fall over time, so the amount of capital exposed to risk changes across horizons.
- These figures motivate the paper's balance-scaled benchmark funds and horizon-specific risk adjustment.

Detailed interpretation: The two figures explain why a simple buy-and-hold corporate-bond benchmark would be inappropriate. A loan's cash-flow risk is not constant over time. Early repayment reduces future exposure, while defaults and recoveries change the timing of realized payoffs. The paper therefore needs benchmark assets that match not only the risk type but also the timing and outstanding balance of the loan cash flows.

How it fits the paper: These figures bridge the data construction and asset-pricing sections. They motivate the use of benchmark funds that are scaled by loan balance and allowed to vary across future horizons.

Economic message. Loans are amortizing private-credit contracts. The benchmark should track the loan's changing exposure, not merely its initial principal amount.

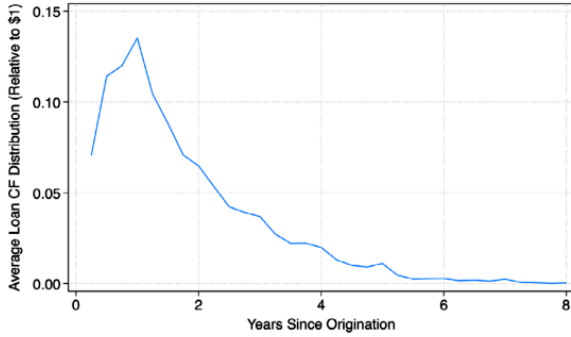
Table 1
Summary Statistics

This table reports summary statistics for the loan sample. *Loan Amt* is the loan facility amount in millions of dollars. *Original Mat* is the original loan maturity in years. *Refinanced* is an indicator equal to one if the loan is prepaid or refinanced before the original maturity. *Years to Refi* is the number of years since origination that a loan is refinanced. *WAL* is the weighted average life of a loan's outstanding principal balance. *Spread Over 3M Tbill* is the loan rate spread over 3M Treasury bill rate at the time of origination. *Default* is an indicator for whether the loan defaults at any point while it is outstanding. *Upfront Fee* is the borrower's fee on the loan paid at origination. *Firm Size* is the borrower's total assets in billions of dollars. *Firm Age* is the number of years a borrower has been active in Compustat truncated at 37 years. *Firm Unrated* indicates whether the firm has an unsecured long-term credit rating from S&P. *Firm Issued Bond* indicates whether the firm has issued a corporate bond in FISD by the loan facility start date. *Lead Lender Retention* is the fraction of the loan retained by the lead lender.

	Mean	SD	Min	P25	P50	P75	Max	N
<i>Loan Amt.</i>	276	585	1	35	113	300	24,000	8,125
<i>Original Mat.</i>	4.988	1.677	0.083	4.000	5.000	6.000	8.000	8,125
<i>Refinanced</i>	0.850	0.357	0.000	1.000	1.000	1.000	1.000	8,125
<i>Years to Refi</i>	1.942	1.273	0.252	0.975	1.595	2.696	7.468	6,918
<i>WAL</i>	1.792	1.138	0.250	0.957	1.500	2.443	8.000	8,125
<i>Spread Over 3M Tbill</i>	329.008	147.889	33.999	240.250	310.564	384.623	1,620.301	8,125
<i>Default</i>	0.045	0.207	0.000	0.000	0.000	0.000	1.000	8,125
<i>Upfront Fee</i>	69.533	77.188	-50.000	25.000	50.000	89.082	1,700.000	8,125
<i>Firm Size</i>	3.296	12.838	0.000	0.248	0.783	2.412	466.001	8,097
<i>Firm Age</i>	15.228	10.992	2.000	6.000	12.000	22.000	38.000	8,125
<i>Firm Unrated</i>	0.469	0.499	0.000	0.000	0.000	1.000	1.000	8,125
<i>Firm Issued Bond</i>	0.438	0.496	0.000	0.000	0.000	1.000	1.000	8,125
<i>Lead Lender Retention</i>	0.514	0.319	0.000	0.248	0.479	0.769	1.000	7,830

sample to term loans because the paydowns on credit lines are unobservable. I only include floating-rate LIBOR loans in the sample as these make up the

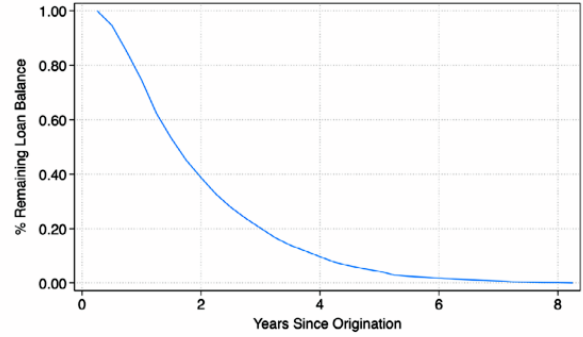
Table I: Summary Statistics. This table summarizes the syndicated term-loan sample. It describes loan size, spreads and fees, maturities, realized weighted-average lives, and borrower or loan characteristics. It shows that the typical loan is large and that realized life and cash-flow timing often differ from the contractual schedule.



Average loan cash flow distribution. This figure plots the quarterly loan portfolio. The cash flow distributions are normalized to a \$1 in

Figure 2: Average Loan Cash Flow

Distribution. This figure plots when realized loan cash flows arrive. It makes clear that loan payoffs are not concentrated only at maturity; a substantial part of the cash-flow stream arrives earlier because of amortization, refinancing, and prepayment.



Loan balance amortization. This figure plots the average outstanding balance since origination. The loan principal balance is normalized to a \$1 in

Figure 3: Loan Balance Amortization.

This figure shows how outstanding balances decline over time. The key reading is that risk exposure shrinks as principal is repaid or prepaid, which is why the paper benchmarks loan cash flows using outstanding-balance weights.

5.3 Table II: Baseline Results; Figure 4: Market Returns to Loan Replicating Portfolio

Read: Table II is the baseline return table, and Figure 4 visualizes the market return series of the loan-replicating benchmark.

- Table II shows that risk-free discounting overstates loan returns because loans load on public-market risk factors.
- After public-market risk adjustment, the loan portfolio still earns positive risk-adjusted profit, around 0.032 dollars per dollar invested, or roughly 177 basis points annualized.
- Figure 4 shows that the benchmark portfolio is itself risky and time varying, so the positive loan result is not a comparison to a risk-free asset.

Detailed interpretation: Table II is the paper’s main quantitative evidence. The central question is not whether loans have high raw spreads, but whether realized loan cash flows are valuable relative to public securities with comparable risk. The positive RAP after risk adjustment implies that the market value of the loan cash-flow stream exceeds the cost of investing in the loans. Figure 4 makes the benchmark tangible: the replicating portfolio has market exposure and fluctuates over time, which clarifies that the estimated loan alpha is not just compensation for obvious public-market risks.

How it fits the paper: This section establishes the gross value of bank lending. Later sections explain where that value comes from and who captures it.

Economic message. Banks earn positive gross risk-adjusted returns on syndicated loan cash flows, but the later expense analysis is needed to determine whether this value becomes shareholder rents.

Baseline Results

This table reports the baseline risk loadings and loan-level risk-adjusted return estimates. Panel A reports the risk loadings for the baseline specification that includes the floating-rate risk-free (Rf) bond, the a_h horizon intercepts for ZCB bonds, Term (10Y Treasury bond returns), BBB (BBB rated corporate bond return), HY (high-yield-rated corporate bond returns), and Stock (CRSP value-weighted stock returns). Rollover loadings are reported in the first row for each factor, with the Rf rollover loading corresponding to a risk-free 3M-treasury bond. Gain loadings are reported by horizon, which corresponds to the number of years since loan origination. The column (1) gain loadings represent the horizon intercepts a_h in equation (6). Panel B reports both the total risk-adjusted return, (RAP), and the annualized risk-adjusted return (ψ). Column (1) reports “risk-free” estimates—that is, cash flows discounted at the risk-free rate. Column (2) reports the baseline estimate that corresponds to the risk factors and loadings in Panel A. Columns (3) and (4) expand the risk factors to include callable bonds, and value/small stocks, respectively. Standard errors are in parentheses and computed using a nonparametric block bootstrap procedure. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Panel A: Baseline Risk Loadings					
	Rf / a_h (1)	Term (2)	BBB (3)	HY (4)	Stock (5)
Rollover Loading	0.9251*** (0.0082)	0.0253*** (0.0077)	0.0515*** (0.0169)	-0.0271*** (0.0071)	0.0144*** (0.0029)
Horizon 1	0.0079*** (0.0002)	0.0028 (0.0062)	-0.0337** (0.0136)	0.0098* (0.0059)	0.0022 (0.0022)
Horizon 2	0.0028*** (0.0002)	0.0136*** (0.0039)	-0.0131 (0.0083)	0.0071* (0.0040)	0.0040*** (0.0010)
Horizon 3	0.0007*** (0.0001)	0.0017 (0.0027)	0.0196*** (0.0046)	-0.0071*** (0.0024)	0.0030*** (0.0007)
Horizon 4	-0.0002** (0.0001)	0.0026 (0.0019)	0.0095*** (0.0030)	0.0001 (0.0012)	0.0009*** (0.0003)
Horizon 5	-0.0001** (0.0001)	0.0009 (0.0010)	0.0062*** (0.0017)	-0.0015* (0.0008)	0.0007*** (0.0002)
Horizon 6	-0.0001*** (0.0000)	0.0007** (0.0004)	0.0008 (0.0006)	0.0007*** (0.0002)	0.0001 (0.0001)
Horizon 7	-0.0000 (0.0001)	-0.0000 (0.0006)	-0.0006 (0.0007)	0.0008*** (0.0002)	-0.0001 (0.0001)
Horizon 8	-0.0000*** (0.0000)	0.0003*** (0.0001)	-0.0003** (0.0001)	0.0002*** (0.0001)	-0.0000 (0.0000)
Observations	259,300	259,300	259,300	259,300	259,300
F-stat	0.000	0.002	0.000	0.000	0.000
R ²	0.991	0.991	0.991	0.991	0.991
Panel B: Risk-Adjusted Return Estimates					
	Risk-Free (1)	Baseline (2)	Callable (3)	Other Stock (4)	
RAP	4.900*** (0.097)	3.176*** (0.133)	3.186*** (0.139)	3.217*** (0.140)	
Annualized RAP (ψ)	2.734*** (0.054)	1.772*** (0.059)	1.777*** (0.078)	1.794*** (0.078)	

Table II: Baseline Results. The table reports the loan portfolio’s risk-adjusted profit under the paper’s benchmark specifications. It shows that risk adjustment reduces the apparent return relative to risk-free discounting, but positive gross RAP remains.

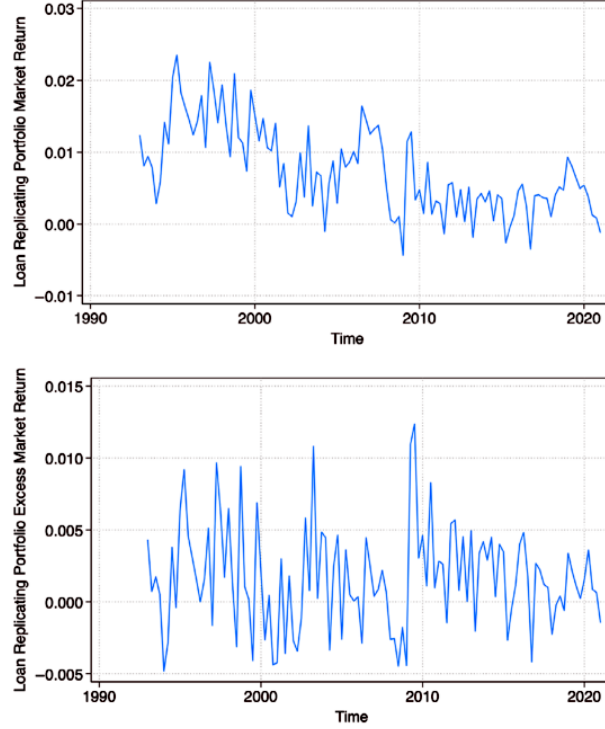


Figure 4: Market Returns to Loan Replicating Portfolio. The figure plots the public-market benchmark portfolio used to replicate loan risk. It highlights that the benchmark is a risky portfolio with its own return variation, not a risk-free discounting rule.

5.4 Table III: Risk-Adjusted Returns and Financial Constraints; Table IV: Screening, Monitoring, and Renegotiation

Read: These tables explain why the average loan alpha is economically meaningful.

- Table III shows that loan RAP is higher for financially constrained borrowers, consistent with the idea that bank services are more valuable when borrowers face financing frictions.
- The use of multiple constraint proxies is important because any single proxy is noisy; repeated sign patterns strengthen the mechanism.
- Table IV connects loan returns to bank services such as screening, monitoring, covenant tightness, and renegotiation capacity.
- The evidence suggests that value comes from costly information production and control rights, not simply from passively holding risky debt.

Detailed interpretation: Table III is a mechanism test. If positive RAP came only from omitted risk, it would not necessarily be concentrated among borrowers that theory predicts should benefit most from relationship lending. The stronger returns for constrained borrowers are consistent with banks being compensated for mitigating asymmetric information, monitoring borrower behavior, and providing credit when public markets are less effective. Table IV sharpens this interpretation by linking returns to features of screening and monitoring. The table suggests that the bank’s active role matters: tighter contracts, costly due diligence, and monitoring-intensive environments are associated with higher returns.

How it fits the paper: These tables move the paper from measurement to economic explanation. They show that loan value is not uniformly distributed; it is strongest where bank intermediation should matter most.

Economic message. Bank loan alpha is a return to solving financing frictions. The evidence is strongest for constrained borrowers and monitoring-intensive loans.

----- Risk-Adjusted Returns and Financial Constraints

This table reports loan-level risk-adjusted returns sorted by the financial constraints of borrowers. I form quartile sorts of loans by the first principal component of the log borrower's total assets (*Firm Size*), the number of years active in Compustat (*Firm Age*), whether the firm has a long-term issuer credit rating (*Firm Unrated*), and whether the firm has issued a corporate bond (*Firm Issued Bond*). The procedure separately estimates risk loadings and mean risk-adjusted returns for loan cash flows within each quartile. Risk-adjusted returns are computed using the baseline specification in Table II. The top panel reports both the total risk-adjusted return (*RAP*), and the annualized risk-adjusted return (ψ) across these sorts. The first column reports the mean risk-adjusted returns across all sorts. "H-L" reports the mean difference in risk-adjusted returns between the bottom and top quartiles. The bottom panel reports the mean characteristics of the loans in each sort. Standard errors are in parentheses and computed using a nonparametric block bootstrap procedure. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Risk-Adjusted Return	Sorts by Financial Constraints					H-L
	Mean	Low	2	3	High	
<i>RAP</i>	3.17*** (0.15)	2.14*** (0.30)	2.43*** (0.37)	3.62*** (0.31)	4.37*** (0.26)	2.23*** (0.39)
<i>Annualized RAP</i> (ψ)	1.77*** (0.08)	1.26*** (0.18)	1.32*** (0.20)	1.98*** (0.17)	2.42*** (0.14)	1.17*** (0.22)
Measure of Constraints	Sorts by Financial Constraints					H-L
	Mean	Low	2	3	High	
<i>Firm Size</i>	3,295.59 (142.67)	9,181.76 (544.78)	2,556.93 (118.42)	1,293.41 (71.97)	477.47 (27.71)	-7,892.86
<i>Firm Age</i>	14.23 (0.12)	22.91 (0.26)	14.04 (0.22)	11.97 (0.20)	8.60 (0.18)	-7.77
<i>Firm Unrated</i>	0.47 (0.01)	0.02 (0.00)	0.19 (0.01)	0.62 (0.01)	0.98 (0.00)	0.96
<i>Firm Issued Bond</i>	0.44 (0.01)	0.97 (0.00)	0.66 (0.01)	0.19 (0.01)	0.00 (0.00)	-0.97

Table III: Risk-Adjusted Returns and Financial Constraints. The table compares loan returns across borrower financial-constraint groups. The key reading is that constrained borrowers generate higher RAP, supporting the hypothesis that banks create more value when borrowers face stronger financing frictions.

Table IV
Screening, Monitoring, and Renegotiation

This table examines loan-level variation in risk-adjusted returns based on whether they are driven by screening, monitoring, or renegotiation. I form sets of loans using *Prob(Reneg)* (probability of loan renegotiation), *Previous Reneg* (whether a loan renegotiated a previous Dealscan loan), *Maturity Length* (the maturity length of the loan), *Industry Fixed Costs* (the average fixed cost of the industry of the borrower), and *EBC Tightness* (the tightness of the firms' earning-based covenants). The procedure separately estimates risk loadings and mean risk-adjusted returns for loan cash flows within sort. Risk-adjusted returns are computed using the baseline specification in Table II. The top panel reports both the total risk-adjusted return (*RAP*), and the annualized risk-adjusted return (ψ) across these sorts. The first column reports the mean risk-adjusted returns across all sorts. "H-L" reports the mean difference in risk-adjusted returns between the bottom and top sorts. The bottom panel reports the mean characteristics of the loans in each sort. Standard errors are in parentheses and computed using a nonparametric block bootstrap procedure. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

	Risk-Adjusted Returns			
	Mean	Low	High	H-L
<i>Prob(Reneg)</i>				
<i>RAP</i>	3.18*** (0.13)	3.45*** (0.19)	2.89*** (0.23)	-0.56** (0.28)
<i>Annualized RAP</i> (ψ)	1.77*** (0.07)	2.09*** (0.11)	1.50*** (0.12)	-0.59*** (0.16)
<i>Prev. Reneg</i>				
<i>RAP</i>	3.18*** (0.13)	3.21*** (0.32)	3.18*** (0.19)	-0.03 (0.30)
<i>Annualized RAP</i> (ψ)	1.77*** (0.07)	1.82*** (0.18)	1.76*** (0.10)	-0.06 (0.17)
<i>Maturity Length</i>				
<i>RAP</i>	3.18*** (0.13)	2.96*** (0.23)	3.24*** (0.35)	0.28 (0.44)
<i>Annualized RAP</i> (ψ)	1.77*** (0.07)	2.35*** (0.18)	1.54*** (0.17)	-0.81*** (0.26)
<i>Industry Fixed Costs</i>				
<i>RAP</i>	3.31*** (0.15)	2.67*** (0.22)	3.92*** (0.17)	1.25*** (0.32)
<i>Annualized RAP</i> (ψ)	1.85*** (0.08)	1.51*** (0.13)	2.17*** (0.10)	0.66*** (0.18)
<i>EBC Tightness</i>				
<i>RAP</i>	3.84*** (0.24)	3.00*** (0.36)	4.64*** (0.27)	1.63*** (0.46)
<i>Annualized RAP</i> (ψ)	2.07*** (0.12)	1.66*** (0.20)	2.44*** (0.14)	0.78*** (0.25)
Measure	Mean Characteristics			H-L
	Mean	Low	High	
<i>Prob(Reneg)</i>	0.73 (0.00)	0.63 (0.00)	0.83 (0.00)	0.20
<i>Previous Reneg</i>	1.75 (0.00)	1.00 (0.00)	2.00 (0.00)	1.00

(Continued)

Table IV: Screening, Monitoring, and Renegotiation. The table relates loan returns to proxies for bank services and contract intensity. It should be read as a test of whether loan alpha is connected to active banking functions such as screening, monitoring, covenant design, and renegotiation.

Table IV—Continued

Measure	Mean Characteristics			H-L
	Mean	Low	High	
<i>Maturity Length</i>	59.85 (0.22)	32.02 (0.33)	80.19 (0.18)	48.19
<i>Industry Fixed Costs</i>	0.29 (0.00)	0.17 (0.00)	0.40 (0.00)	0.23
<i>EBC Tightness</i>	0.19 (0.00)	0.04 (0.00)	0.32 (0.00)	0.29

Table IV continued. The continuation adds further specifications and controls. The purpose is to test whether the financial-constraint and monitoring interpretation survives after accounting for observable contract and borrower characteristics.

5.5 Figure 5: Histogram of Bank-Level Alphas; Table V: Bank-Level Variation

Read: These visuals ask whether loan value differs across banks, not just across loans.

- Figure 5 shows meaningful dispersion in bank-level alphas, implying that average RAP masks heterogeneity across lenders.
- Table V shows that banks with portfolios tilted toward constrained borrowers earn higher annualized RAP.
- The results suggest that borrower mix, bank expertise, relationships, and lending capacity all matter for bank-level value creation.

Detailed interpretation: The bank-level analysis matters because the loan-level results could otherwise be interpreted as purely borrower-driven. Figure 5 indicates that banks differ in their gross value creation. Table V links those differences to the kinds of borrowers banks serve. This is consistent with the idea that banks have scarce lending capacity and heterogeneous ability to screen, monitor, and manage private credit. It also helps explain why competition may not eliminate gross returns immediately: if lending technology, relationship capital, and skilled labor are scarce, returns can persist across banks.

How it fits the paper: This subsection connects the loan-level findings to the organization of the banking industry. It shows that value is partly attached to bank balance sheets and lending technologies.

Economic message. Bank alpha is partly a return to scarce lending capacity and specialized private-credit expertise.

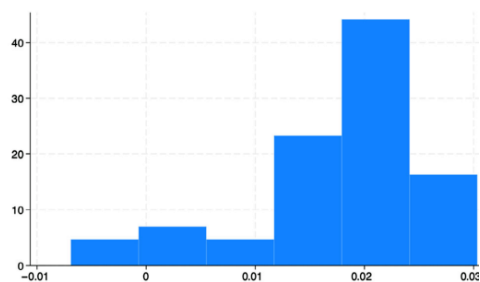


Figure 5: Histogram of Bank-Level Alphas. The figure shows the distribution of bank-level alphas. Its main message is that banks differ substantially in gross lending returns, rather than all banks earning the same average alpha.

Bank-Level Variation

This table examines bank-level determinants of risk-adjusted returns. The dependent variable is the bank-level annualized risk-adjusted returns, which are computed by summing loan cash flows into value-weighted portfolios based on the fraction of the loans retained at origination and estimating risk loading separately for each bank using equation (8). *Portfolio PC1(Constraints)* is the average level of firms' financial constraints *PC1(Constraints)* in a given bank's portfolio. *Bank Size* is the log of the total assets of the bank. *Commercial Loans* is the fraction of commercial loans of total assets on the bank's balance sheet. *Wholesale Funding* is the fraction of commercial paper and certificates of deposit used by the bank. *Demand Deposits* is the fraction of demand deposits to assets of a bank. *Equity Ratio* is the book equity ratio of a bank. Independent variables in this table are standardized to have a mean of zero and a standard deviation of one. Standard errors are in parentheses and clustered at the bank level. $p < 0.10$, $**p < 0.05$, $***p < 0.01$.

	(1)	(2)	(3)	(4)	(5)	(6)
<i>Portfolio PC1(Constraints)</i>	1.187*** (0.253)	1.592*** (0.388)	1.188*** (0.364)	1.143*** (0.352)	1.358*** (0.350)	
<i>Bank Size</i>	-0.358* (0.209)	-0.587** (0.215)	-0.079 (0.237)	-0.310 (0.217)	-0.463*** (0.153)	
<i>Syn. Port Size</i>	0.328 (0.206)	0.473** (0.189)	0.185 (0.204)	0.406* (0.200)	0.460** (0.170)	
<i>Wholesale Funding</i>		-0.445** (0.191)				
<i>Wholesale Funding × Portfolio PC1(Constraints)</i>		-0.214 (0.463)				
<i>Demand Deposits</i>			-0.144 (0.155)			
<i>Demand Deposits × Portfolio PC1(Constraints)</i>			-1.258** (0.483)			
<i>Equity Ratio</i>				0.308** (0.124)		
<i>Equity Ratio × Portfolio PC1(Constraints)</i>				0.131 (0.596)		
<i>Commercial Loans</i>						-0.295** (0.106)
<i>Commercial Loans × Portfolio PC1(Constraints)</i>						-0.641 (0.483)
Constant	1.760*** (0.136)	1.949*** (0.133)	2.031*** (0.142)	2.015*** (0.142)	1.932*** (0.125)	2.026*** (0.144)
Observations	31	31	31	31	31	31
R ²	0.000	0.215	0.504	0.451	0.422	0.393

Table V: Bank-Level Variation. The table relates bank-level returns to bank portfolio composition and other bank characteristics. It shows that banks lending more to constrained borrowers tend to earn higher gross RAP.

5.6 Table VI: Bank-Level Persistence; Table VII: Commercial Loan Expenses

Read: These tables separate value creation from shareholder rents.

- Table VI shows persistence in bank-level RAP and dollar value added, which is consistent with durable bank skill, relationships, or organizational capital.
- Table VII shows that lending expenses are close to the gross annualized RAP, especially when staff compensation and noninterest expenses are included.
- The two tables together imply that banks create gross value, but much of that value is paid to the inputs needed to produce lending services.

Detailed interpretation: Persistence in Table VI argues against the view that positive bank value-added is merely sampling noise. Some banks appear to have durable advantages in generating loan cash-flow value. Table VII then changes the distributional interpretation of the result. Positive gross RAP does not mean bank shareholders earn large abnormal profits. Commercial lending requires costly personnel, monitoring systems, credit administration, and relationship-management infrastructure. These costs absorb most of the gross value. The distinction between gross and net value is crucial for welfare: lending can be socially valuable even when shareholders earn near-zero net alpha.

How it fits the paper: This subsection clarifies who captures the value of bank lending. Gross value exists, but it is largely paid to labor and organizational inputs rather than becoming pure shareholder rent.

Economic message. Banks create value through costly intermediation. Net shareholder alpha can be near zero even when lending generates positive gross surplus.

Bank-Level Persistence					
This table examines persistence in bank-level risk-adjusted returns. Risk-adjusted returns at the bank-time level are estimated using equation (7) using the baseline set of risk factors with risk loadings estimated separately for each bank's portfolios. This table tests for persistence by regressing risk-adjusted returns on lagged risk-adjusted returns where <i>L4</i> indicates a four-quarter or one-year lag. Panel A reports results using the risk-adjusted return per dollar of loan (<i>RAP</i>), and Panel B reports results using the <i>Dollar Value-Added</i> , which is the product of the loan portfolio amount (in millions) in a given quarter and the risk-adjusted return per dollar of loan in that quarter. All specifications include quarter-time fixed effects. Standard errors clustered by bank are in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.					
Panel A: RAP Persistence					
	<i>RAP</i>				
	(1)	(2)	(3)	(4)	(5)
Constant	0.032*** (0.002)	0.029*** (0.002)	0.030*** (0.003)	0.030*** (0.003)	0.032*** (0.003)
<i>L4.RAP</i>		0.101*** (0.032)			
<i>L8.RAP</i>			0.080** (0.033)		
<i>L12.RAP</i>				0.126*** (0.032)	
<i>L16.RAP</i>					0.089*** (0.024)
Observations	2,755	2,451	2,304	2,146	2,000
Time FE	No	Yes	Yes	Yes	Yes
R^2	0.000	0.144	0.147	0.150	0.138
Panel B: Total Value-Added Persistence					
	<i>Dollar Value-Added</i>				
	(1)	(2)	(3)	(4)	(5)
Constant	18.727*** (4.160)	12.785*** (2.286)	13.177*** (2.420)	15.241*** (2.871)	18.687*** (3.635)
<i>L4.Dollar Value-Added</i>		0.473*** (0.121)			
<i>L8.Dollar Value-Added</i>			0.582*** (0.093)		
<i>L12.Dollar Value-Added</i>				0.603*** (0.129)	
<i>L16.Dollar Value-Added</i>					0.517*** (0.107)
Observations	2,755	2,451	2,304	2,146	2,000
Time FE	No	Yes	Yes	Yes	Yes
R^2	0.000	0.302	0.325	0.303	0.243

Table VI: Bank-Level Persistence. The table tests whether banks with higher value-added in one period continue to generate value in later periods. Persistence supports the interpretation that bank lending value reflects durable bank capabilities rather than transitory luck.

Commercial Loan Expenses

This table estimates the expense ratios for commercial lending using the hedonic regression approach of Hanson et al. (2015) on FRY-9C banks from 1994 to 2014. Using equation (10), I regress the total noninterest expense of banks on commercial loans, real estate loans, consumer loans, trading assets, other assets (excluding cash and securities), transaction deposits, savings deposits, and foreign deposits, leaving equity and wholesale funding as the omitted category. The expense ratio specific to commercial lending is taken as the predicted value of a bank that invests only in commercial loans using only equity and wholesale funding. Below I report the estimated total noninterest expense for commercial lending as well as the estimated compensation ratio and other noncompensation expenses of commercial lending. The panels report the estimated expense ratios for the full sample of FRY-9C banks as well as for banks active in the syndicated loan sample, and banks with zero trading assets. Following Hanson et al. (2015), the estimates are obtained using Fama-Macbeth regressions, and Newey-West (1) standard errors are reported in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

	Total Noninterest Expense	Staff Comp.	Noncomp
Full BHC Sample			
<i>Expense(CI_Loans)</i>	1.818***	1.126***	0.692***
<i>CI_Loans</i>	(0.100)	(0.044)	(0.067)
Syndicated Lenders Only			
<i>Expense(CI_Loans)</i>	1.653***	1.239***	0.414**
<i>CI_Loans</i>	(0.235)	(0.139)	(0.198)

Table VII presents the estimates from this exercise.³⁹ Using the full sample of bank holding companies, the estimated total noninterest expense for com-

Table VII: Commercial Loan Expenses. The table compares gross lending returns with commercial lending expenses. It shows that the costs of staff, monitoring, administration, and other noninterest inputs are large enough to absorb most of the gross RAP.

5.7 Table VIII: Valuing Corporate Bond Cash Flows

Read: Table VIII is the paper’s placebo test for the risk-adjustment methodology.

- Corporate bonds are useful because their market prices are observed, so a valid benchmark should not create large artificial alpha for bond cash flows.
- Applying the same methodology to public corporate bonds produces near-zero risk-adjusted returns across subsamples.
- This makes it less likely that positive loan RAP is a mechanical artifact of the benchmark construction.

Detailed interpretation: Table VIII is central for credibility. If the benchmark method generated positive RAP for any debt-like cash flow, then the loan results would be weak evidence. The near-zero bond result shows that the method works on public securities whose market values are known. This validation supports the paper’s interpretation that positive loan RAP is specific to private bank lending and the services bundled with it, rather than a generic mispricing of credit cash flows.

How it fits the paper: The placebo test anchors the asset-pricing design. It demonstrates that the benchmark can price public bond cash flows before being used to value private loan cash flows.

Economic message. Positive loan RAP is not a generic debt-pricing artifact. It is tied to private bank loan cash flows and the intermediary services embedded in them.

Valuing Corporate Bond Cash Flows

This table repeats the baseline estimation exercise using corporate bond cash flows for which market prices are available. Panel A reports the risk loadings for the full sample of corporate bond cash flows as in Table II. Panel B reports the risk-free and annualized risk-adjusted returns, which are labeled *Rf Ret.* and *RAP Ann.*, respectively. Columns (1) and (2) include bonds across all maturities, columns (3) and (4) include bonds with less than five years of maturity, and columns (5) and (6) include bonds with more than five years of maturity. The table reports estimates for the full sample of corporate bond cash flows as well as subsamples of corporate bonds by investment grade (IG), high-yield (HY), low offering spread (below the sample median), and high offering spread (above the sample median). Standard errors are in parentheses and computed using a nonparametric block bootstrap procedure. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Panel A: Full-Sample Risk Loadings					
	Rf / a_h (1)	Term (2)	BBB (3)	HY (4)	Stock (5)
Rollover Loading	0.9091*** (0.0166)	-0.0174 (0.0173)	0.1009*** (0.0370)	-0.0392*** (0.0150)	0.0085 (0.0058)
Horizon 1	0.0061*** (0.0002)	0.0864*** (0.0150)	-0.0680** (0.0317)	0.0571*** (0.0134)	0.0222*** (0.0044)
Horizon 2	0.0040*** (0.0003)	0.0533*** (0.0121)	-0.0220 (0.0229)	0.0286** (0.0118)	0.0171*** (0.0040)
Horizon 3	0.0022*** (0.0004)	0.0462*** (0.0130)	-0.0476** (0.0209)	0.0389*** (0.0103)	0.0157*** (0.0028)
Horizon 4	0.0008** (0.0004)	0.0374** (0.0181)	-0.0320 (0.0234)	0.0223*** (0.0083)	0.0146*** (0.0021)
Horizon 5	0.0009*** (0.0003)	0.0442*** (0.0140)	-0.0234 (0.0174)	0.0217*** (0.0059)	0.0118*** (0.0017)
Horizon 6	0.0005** (0.0002)	0.0286** (0.0132)	-0.0013 (0.0191)	-0.0021 (0.0063)	0.0092*** (0.0014)
Horizon 7	-0.0001 (0.0001)	0.0365*** (0.0078)	-0.0092 (0.0097)	0.0083** (0.0038)	0.0047*** (0.0013)
Horizon 8	-0.0007*** (0.0001)	0.0284*** (0.0076)	-0.0003 (0.0096)	-0.0045* (0.0025)	0.0048*** (0.0009)
Observations	81,468	81,468	81,468	81,468	81,468
F-stat p -value	0.000	0.000	0.165	0.000	0.000
R ²	0.967	0.967	0.967	0.967	0.967

Panel B: Risk-Adjusted Returns Estimates						
	All Mat		Low Mat		High Mat	
	(1) $Rf Ret.$	(2) $RAP Ann.$	(3) $Rf Ret.$	(4) $RAP Ann.$	(5) $Rf Ret.$	(6) $RAP Ann.$
Full Sample	2.769*** (0.077)	0.015 (0.102)	1.718*** (0.103)	-0.146 (0.146)	3.160*** (0.094)	0.060 (0.124)
IG Bonds	1.153*** (0.055)	-0.149 (0.093)	1.056*** (0.077)	-0.239 (0.151)	1.255*** (0.076)	-0.037 (0.133)
HY Bonds	4.043*** (0.115)	-0.121 (0.138)	5.082*** (0.393)	0.176 (0.434)	3.953*** (0.120)	-0.157 (0.156)

(Continued)

Table VIII: Valuing Corporate Bond Cash Flows, part 1. This table applies the loan-valuation method to public corporate bond cash flows. The near-zero risk-adjusted returns indicate that the benchmark procedure does not mechanically generate positive alpha.

Panel B: Risk-Adjusted Returns Estimates						
	All Mat		Low Mat		High Mat	
	(1) <i>Rf Ret.</i>	(2) <i>RAP Ann.</i>	(3) <i>Rf Ret.</i>	(4) <i>RAP Ann.</i>	(5) <i>Rf Ret.</i>	(6) <i>RAP Ann.</i>
Low Spread	0.468*** (0.052)	0.025 (0.094)	0.444*** (0.062)	0.015 (0.152)	0.495*** (0.085)	0.039 (0.162)
High Spread	3.426*** (0.092)	0.113 (0.103)	2.680*** (0.163)	−0.022 (0.221)	3.610*** (0.104)	0.073 (0.143)

HY and longer maturity, low offering spread and low maturity, etc). I find that

Table VIII continued. The continuation reports additional bond subsamples and specifications. The same near-zero result reinforces that the loan alpha is specific to private bank lending rather than public debt cash flows.

6 Short summary

- Loan cash flows generate about 177 bps in annualized gross risk-adjusted returns.
- Returns are higher for financially constrained borrowers and monitoring-intensive loans.
- Bank-level alphas are persistent, especially in dollar value-added terms.
- Commercial lending expenses roughly exhaust gross RAP, leaving shareholders near zero net alpha.

- Corporate bond placebo tests support the risk-adjustment methodology.

7 Conclusion

Flanagan shows that bank lending creates measurable gross value. Realized syndicated loan cash flows outperform public-market benchmarks with similar risk, supporting classic theories in which banks mitigate borrower information and incentive problems. The returns are highest where borrower frictions are likely most severe, suggesting that banks matter because they transform costly screening and monitoring into credit that public markets cannot perfectly replicate.

However, the positive gross RAP does not imply shareholder rents. Commercial lending expenses, especially staff compensation, absorb most of the gross return. RAP measures the lender-side value only; borrowers may also receive surplus through lower financing costs or access to credit, so lender RAP should be interpreted as a lower bound on total surplus generated by bank lending.

The policy implication is that high loan returns should not automatically be interpreted as excessive risk-taking. They may instead compensate banks for costly information production and monitoring. At the same time, bank financing frictions can raise required returns and restrict credit to constrained borrowers. The main limitations are that the sample focuses on syndicated loans to public firms and that the methodology relies on benchmark spanning of priced risks.